nature human behaviour

Matters arising

https://doi.org/10.1038/s41562-025-02224-3

High variability in LLMs' analogical reasoning

Received: 21 October 2024

Accepted: 24 April 2025

Published online: 04 June 2025

Check for updates

Andrea Gregor de Varda[1,2], Chiara Saponaro[1] & Marco Marelli[1]

ARISING FROM T. Webb et al. *Nature Human Behaviour* https://doi.org/10.1038/s41562-023-01659-w (2025)

In a recent study, Webb, Holyoak and Lu[1] (henceforth WHL) demonstrated that a large language model (GPT-3, text-davinci-003) could match or even exceed human performance across several analogical reasoning tasks. This result led to the compelling conclusion that LLMs such as GPT-3 possess an emergent ability to reason by analogy. However, the findings were based on a single, proprietary model for which the releasing company provided limited public details and progressively restricted access to the internal probability distributions. Furthermore, text-davinci-003 was deprecated on 4 January 2024, and is no longer available through the OpenAI API. This poses a challenge to replicability in two ways. First, the lack of open access to the model and its recent deprecation make it difficult—if not impossible—for other researchers to verify or build upon the findings. Second, relying on a single model leaves open the question of whether the results can be extended to LLMs as a broader class of objects of scientific investigation. Without testing a diverse range of models, it is unclear whether the observed behaviours are specific to GPT-3 or represent a general property of comparable contemporary LLMs. Replicating experimental results based on proprietary models with public alternatives is thus crucial to ensure that the findings can be reproduced in the future[2], generalized to new model instances, and, more generally, to adhere to transparency principles that are of paramount importance in scientific research[3].

In this replication effort, conducted in the context of a collaboration between Nature Human Behaviour and the Institute for Replication (I4R)[4], we collected new data from 26 high-performance LLMs. In particular, we tested models from the Phi[5], Gemma[6], Llama[7] and Mistral[8] model families; our model sample covers both 'base' LLMs trained with a self-supervised next-token prediction objective, and 'instruct' and 'chat' models which are further optimized with supervised learning to follow instructions. All the LLMs we considered are open weights and released under permissive licences; they are large Transformer-based models (up to ~70 B parameters) trained on huge corpora of natural language data (up to ~15 T tokens; see Supplementary Information 1.3 for additional details). We collected model responses for the stimuli in Experiments 1, 2 and 4 of WHL's paper (we did not replicate Experiments 3 and 5 since the comparison between model and human responses was conducted in qualitative terms and not in the form of hypothesis testing). In those experiments, GPT-3 was tested on its ability to solve analogical problems presented in the form of text-based matrix reasoning problems (Experiment 1, 'Digit matrices'), letter string analogies (Experiment 2, 'Letter string') and story analogies (Experiment 4, 'Story analogies'; see Supplementary Information 1.2 for an overview of the tasks). Across the experiments, the models were evaluated by generating answers directly; in Experiment 1, the model was additionally tested by selecting an answer from a set of alternatives on the basis of the log-probabilities assigned to input tokens (see Supplementary Information 2.1). We aimed for a statistical power of at least 90%, as calculated with 0.8 of the original effect sizes. To this end, we evaluated the models on the original stimuli employed by WHL in Experiments 1 and 2; in Experiment 4, we created seven new story items to achieve the desired statistical power (see Supplementary Information 1.1). With the intention of evaluating WHL's main claims (see Supplementary Information 1), we compared the models' performance against human-level accuracy in Experiments 1 and 2, and against chance-level accuracy in Experiment 4.

Our results highlight high variability in the LLMs' analogical reasoning performance, both in terms of intermodel differences in the response profiles and inter-task differences in average model performance. In the Digit matrices task, all models performed above chance level, and most models achieved superhuman generative accuracy in solving text-based matrix reasoning problems (Fig. 1a; see Supplementary Information 2.1 Table 2 for the multiple-choice results). In contrast, their performance dropped markedly in the Letter string task, where all the models we tested performed above chance level (chance-level accuracy in the generative tasks is close to zero due to the large space of possible responses) but consistently worse than human reasoners (Fig. 1b). Finally, in the Story analogies condition, performance was strikingly variable: while some models achieved very high accuracy scores (up to 0.85, $P < 0.0001$), many LLMs performed significantly below chance (Fig. 1c). Notably, all the models that displayed above-chance reasoning abilities in this condition were instruction tuned. Our results partially align with the findings reported by WHL in that at least some models consistently performed above chance level across all three analogical

[1]Department of Psychology, University of Milano - Bicocca, Milan, Italy. [2]Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA, USA. e-mail: devar_ag@mit.edu

https://doi.org/10.1038/s41562-025-02224-3

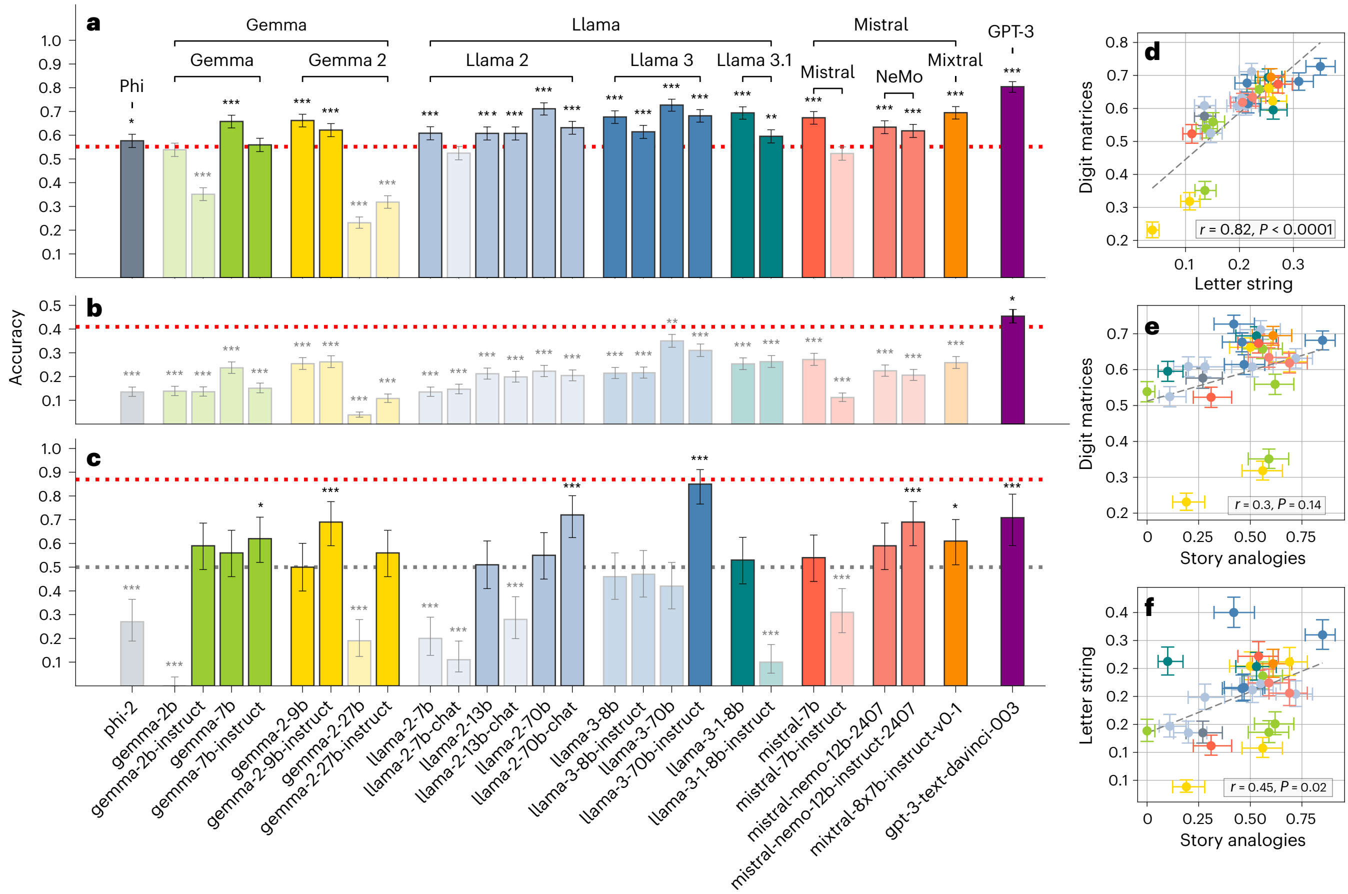

Fig. 1 | Performance of LLMs on analogical reasoning tasks. a, Generative performance in the Digit matrices task. **b**, Generative performance in the Letter string task. **c**, Accuracy in the Story analogies condition. Across the three subplots, the dotted lines report human (red; **a**–**c**) and chance-level (grey, **c**) accuracy. Note that accuracy from human participants and GPT-3 in subplot **c** is calculated on the original problems from WHL. The asterisks indicate two-tailed statistical significance (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$). Their colour indicates whether performance is significantly above (black) or below (grey) human accuracy (**a**,**b**) and chance level (**c**). Models with subhuman (**a**,**b**) or below-chance (**c**) accuracy have lower opacity. The error bars indicate 95% binomial confidence intervals for the average performance across different problems. **d**–**f**, Model-wise correlation of accuracy scores across tasks. The dashed grey lines indicate the best linear fit. The error bars indicate 95% binomial confidence intervals along the relevant axes.

reasoning tasks. Yet, the stark inter-model and inter-task differences indicate that analogical reasoning is not a uniform trait of LLMs but is contingent on both the models' training regimen and the task demands that are specific to each analogy condition.

A comparative analysis of the models' performance revealed a strong, significant correlation between their accuracy scores in the Letter string and the Digit matrices tasks (Fig. 1d). This strong correlation suggests that these forms of analogical reasoning likely tap into similar abilities—arguably, the identification of relations among symbolic units. Conversely, accuracy scores in the Story analogies condition only exhibited a weak-to-moderate correlation with the two other tasks (Fig. 1e,f), pointing to differences in the reasoning mechanisms at play. Specifically, the Story analogies may additionally require an understanding of more complex, high-level relationships between causal structures in naturalistic narratives.

Traditional experiments in the behavioural sciences often assume that human participants can be treated as a relatively homogeneous and stable sample. LLMs, on the other hand, constitute a fundamentally heterogeneous category[9]: they vary greatly in the amount of pre-training data, number of parameters, training objectives and the presence of human feedback during training. This raises a critical question: under what conditions should we assert that LLMs, as a natural class of objects of scientific investigation, exhibit a target behaviour? The issue of generalizability is rarely discussed in cognitively oriented research into LLMs. One theoretical possibility is that a behaviour documented for a given LLM should be expected to extend to all other model instances; however, this is unrealistic, as some models are simply worse than others, whether due to size, training regimen, or architectural differences. A more plausible assumption is that a positive result in one model will generalize to LLMs that are at least as capable, typically those with similar or lower perplexity, larger size, or stronger overall performance. While all the models we tested are smaller than text-davinci-003 (175 B parameters; see Supplementary Information 1.3), some achieve comparable, if not better, performance on standardized benchmarks[10]. Thus, even the notion that behaviours will generalize to similarly performing models is not always warranted. Instead, claims of generalizability should be made explicit and rest on empirical evidence.

The evidence we presented underscores the need to define clear standards for the scientific investigation of behavioural phenomena in LLMs. Proprietary models may display interesting behaviours sooner than their open-weights counterparts, but the higher transparency associated with the latter represents a crucial feature for rigorous scientific inquiry. Moving forward, commercial LLMs can provide a valuable starting point for exploring complex behaviours, but replicating findings with open-weights models is crucial to abide by transparency principles that uphold the standards of open, reproducible science.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

References

1. Webb, T., Holyoak, K. J. & Lu, H. Emergent analogical reasoning in large language models. *Nat. Hum. Behav.* **7**, 1526–1541 (2023).
2. Hu, J. & Levy, R. P. Prompting is not a substitute for probability measurements in large language models. In *Proc. 2023 Conference on Empirical Methods in Natural Language Processing* (eds Bouamor, H. et al.) 5040–5060 (Association for Computational Linguistics, 2023).
3. Binz, M. et al. How should the advancement of large language models affect the practice of science? *Proc. Natl Acad. Sci.* **122**, e2401227121 (2025).
4. Brodeur, A., Dreber, A., Hoces de la Guardia, F. & Miguel, E. Reproduction and replication at scale. *Nat. Hum. Behav.* **8**, 2–3 (2024).
5. Gunasekar, S. et al. Textbooks are all you need. Preprint at https://doi.org/10.48550/arXiv.2306.11644 (2023).
6. Gemma Team. Gemma: open models based on Gemini research and technology. Preprint at https://doi.org/10.48550/arXiv.2403.08295 (2024).
7. Touvron, H. et al. Llama 2: open foundation and fine-tuned chat models. Preprint at https://doi.org/10.48550/arXiv.2307.09288 (2023).
8. Jiang, A. Q. et al. Mistral 7B. Preprint at https://doi.org/10.48550/arXiv.2310.06825 (2023).
9. Apidianaki, M., Fourtassi, A. & Padó, S. Language Learning, Representation, and Processing in Humans and Machines: Introduction to the Special Issue. *Comput. Linguist.* **50**, 1201–1210 (2024).
10. Grattafiori, A. et al. The Llama 3 herd of models. Preprint at https://doi.org/10.48550/arXiv.2407.21783 (2024).

Author contributions

A.G.d.V. conceptualized the project, developed the methodology, conducted investigation, curated data, performed visualization, formal analysis and validation, developed software and wrote the original manuscript draft. C.S. performed investigation and data curation, and reviewed and edited the manuscript. M.M. reviewed and edited the manuscript, and supervised and administered the project.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41562-025-02224-3.

Correspondence and requests for materials should be addressed to Andrea Gregor de Varda.

Peer review information *Nature Human Behaviour* thanks Anna Ivanova and the other, anonymous reviewer(s) for their contribution to the peer review of this work.

nature portfolio

Corresponding author(s): Andrea Gregor de Varda

Last updated by author(s): Feb 27, 2025

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Model responses were collected through the Predibase API (https://predibase.com/).
Data analysis	Data analysis was performed with Python 3.9.7 on a computer with Ubuntu 22.04.4 LTS using the code released by the authors of the article we replicated (Webb, Holyoak, and Lu, 2023 Nat. Hum. Behav.). The code was adapted to collect responses from the Predibase API, since Webb, Holyoak, and Lu used the OpenAI API. The code makes heavy use of numpy (v. 1.26.4) and pandas (v. 1.5.1).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Data is available on OSF (https://osf.io/682fg/).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	We did not collect data from human participants.
Reporting on race, ethnicity, or other socially relevant groupings	We did not collect data from human participants.
Population characteristics	We did not collect data from human participants.
Recruitment	We did not collect data from human participants.
Ethics oversight	We did not collect data from human participants.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	The study analyzes responses from 26 open-weights LLMs to analogical tasks to replicate the results reported by Webb, Holyoak, and Lu, which were based on GPT-3.
Research sample	We collected responses from 26 high-performance LLMs. In particular, we tested models from the Phi , Gemma, Llama, and Mistral model families.
Sampling strategy	The number of model tested was chosen to include prominent models available at the time of experimentation (July - October 2024). The number of items was determined through simulation-based power analysis.
Data collection	Model responses were collected through the Predibase API. The models were tested with temperature equal to 0.
Timing	July - October 2024.
Data exclusions	No data was excluded from the analysis.
Non-participation	N.A.
Randomization	Models were not allocated to experimental groups.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.